

ASSIGNMENT – 1

TESLA CAR FUNCTION

Tesla's self-driving system, known as Full Self-Driving (FSD), is designed to enable autonomous driving using advanced artificial intelligence and deep learning techniques. The system relies on multiple cameras placed around the vehicle to provide a 360-degree view of its surroundings. Using computer vision, it can detect lanes, traffic signals, pedestrians, and other vehicles in real time. Tesla's Autopilot feature assists with steering, braking, and acceleration, making driving safer and more convenient. The system continuously analyzes data from sensors and cameras to make quick driving decisions. Tesla collects vast amounts of real-world data from its fleet of vehicles, which is used to train its neural network models. These models are trained using powerful computing systems like GPUs and the Dojo supercomputer. Updates are delivered through Over-the-Air (OTA) updates, allowing the system to improve without physical changes. Over time, Tesla's self-driving technology becomes more accurate and efficient through continuous learning.

Training of Deep Learning Models in Tesla

1. Data Collection

- Tesla collects real-world driving data from millions of vehicles.
- Cameras installed in cars capture road conditions and surroundings.
- The collected data is stored and used to train AI models.

2. Data Annotation

- The collected data is labeled to identify objects such as cars, pedestrians, and signals.
- Both human annotators and automated tools are used for labeling.

3. Model Training

- Tesla uses deep neural networks such as Convolutional Neural Networks (CNNs).
- Training is performed using high-performance systems like GPUs and the Dojo supercomputer.

4. Simulation Testing

- Models are tested in simulated environments before real-world deployment.
- Different scenarios like accidents, weather conditions, and rare situations are tested.

5. Deployment

- Trained models are deployed to Tesla cars using OTA updates.
- Vehicles receive improvements without needing physical upgrades.

6. Continuous Learning

- Tesla uses a feedback loop where cars send new data back to improve models.
- This process is known as fleet learning and helps the system improve continuously.